Michael A. Morrison

Software Engineering and Scientific Computing

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Software developer and technical lead building ML and multi-agent systems for scientific computing and HPC.

PROFESSIONAL EXPERIENCE

Lawrence Livermore National Laboratory May 2022 - Present

Software Developer | Co-development lead: MADA / MADA-tools

- Led a machine learning joint working group spanning multiple national laboratories for 2.5 years, enabling new cross-lab collaborations through more effective communication.
- Co-led development of MADA and MADA-tools, enabling users to build and demonstrate novel simulation workflows through the platform. MADA-tools provided a Python library of Model Context Protocol (MCP) servers for HPC scheduling, simulation, and surrogate modeling.
- Led development of a multi-person AML effort for 3 years, integrating surrogate modeling with robust simulation workflows. Contributed to the Advanced Machine Learning portfolio and Genesis Mission through ML research for HPC and model integration into complex simulation and modeling workflows.
- Refactored MADA's Python engine into configurable orchestration strategies and implemented Magentic coordination. Added agent validation, isolated request histories, and regression tests across CLI, Gradio, and API interfaces.
- Researched ML for MIR (Material Interface Reconstruction) in Julia and Python, using statistical clustering to tune datasets and CNNs trained on discretized patches for volume-fraction visualization.

AT&T Sep 2020 - Apr 2022

Specialist Software Design Engineer

- Developed and maintained cross-platform desktop applications for high-fidelity simulations and large-scale network analysis.
- Developed concurrent parametric scripts and supported software integration; led quality assurance automation efforts within the development team.

Sacramento Municipal Utility District (SMUD) Feb 2019 - Nov 2019

STEM Student Intern, Digital & UX

- Developed version 1.0 websites for California Mobility Center (now Cal EPIC) and Sacramento Power Academy.
- Built SMUD's Why TOD scrolling experience with jQuery, GSAP, and ScrollMagic to explain time-of-day electricity rates.
- Implemented SMUD.org features and SharePoint intranet improvements within a Scrum team using Sitecore, Visual Studio, and Atlassian tools.

TECHNICAL SKILLS

Languages Python, Julia, C++, C#, JavaScript, SQL

Machine learning and agents TensorFlow/Keras, Torch, Flux.jl, Microsoft Agent Framework, MCP, NumPy, pandas

Web React, Node.js, Express, MongoDB, Gradio, FastAPI, HTML/CSS

Tools Git, Linux, pytest, Visual Studio, Jira, Sitecore, SharePoint

EDUCATION

California State University, Sacramento Dec 2019

B.S. Computer Science, cum laude

Folsom Lake College

A.S. Computer Science; A.A. Interdisciplinary Studies, Math and Science